

Shantanu Shastri

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EDUCATION

Carnegie Mellon University	December 2026
Master of Information Systems Management	GPA: 3.85
Manipal Institute of Technology	August 2022
Bachelor of Electronics and Communications Engineering	GPA: 3.99
<i>Relevant Coursework:</i> Machine Learning, NLP, Generative AI, Data Structures, Algorithms	

SKILLS

Languages: Python, C++, Go, Java, TypeScript, SQL, Bash/Shell
ML & AI: RAG Systems, NLP, Ranking & Search Relevance, Generative AI, Evaluation Systems
Frameworks: PyTorch, TensorFlow, Scikit-learn, XGBoost, Spark MLlib, Pandas
MLOps & Infrastructure: AI Infrastructure, Scalable Data Pipelines, CI/CD, Observability, Milvus, Neo4j

EXPERIENCE

BlackRock	January 2022 – August 2026
<i>AI Engineer Internship</i>	June 2026 – August 2026

- **RAG & NLP Features:** Designed and productionized ML models for Retrieval-Augmented Generation (RAG) and NLP features, achieving 92% precision in classifying and translating complex financial queries.
- **Search & Data Pipelines:** Built scalable data pipelines and hybrid vector-graph infrastructure (Milvus, Neo4j) to effectively drive intelligent search relevance and ranking algorithms across 3.5M records.
- **Infrastructure & Observability:** Partnered across teams to develop robust AI infrastructure for inference and deployment, implementing high-quality observability frameworks to monitor AI model quality and drift.

<i>Software Developer 2 - Associate</i>	January 2025 – July 2025
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- **Generative AI & Datasets:** Collaborated with AI researchers to iterate on generative AI models, successfully fine-tuning Llama-3 (LoRA) on a 40K+ training dataset corpus to achieve 94% Exact Match.
- **Evaluation Systems:** Developed scalable evaluation systems and precise metrics to closely monitor training datasets and model architectures, ensuring high-quality, ML-driven capabilities in production environments.

<i>Software Developer 1 - Analyst</i>	July 2022 – December 2024
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- **Applied ML (XGBoost):** Leveraged Python, Scikit-learn, and XGBoost to design and deploy predictive models, applying an end-to-end ML mindset to achieve 88% precision for mission-critical services.
- **MLOps & CI/CD Automation:** Engineered CI/CD automation for ML workflows via Azure DevOps, focusing heavily on MLOps best practices to seamlessly optimize systems for training and inference.
- **Mentorship & Excellence:** Actively mentored junior engineers and contributed to a culture of technical excellence, rigorously reviewing code for scalable ML data processing frameworks.

Shunya	June 2020 – September 2020
<i>Data Science Intern</i>	

- **Creative ML & Annotation:** Engineered PyTorch CV workflows on Linux for creative evaluation, building data annotation pipelines that reliably automated answer sheet grading by 90%.
- **Product Mindset:** Demonstrated a strong product mindset by tying technical ML clustering work directly to user outcomes, dynamically recommending targeted practice to drive immediate business impact.

PROJECTS

FinTech Recommendation & Ranking System

- **Applied ML Infrastructure:** Developed a comprehensive ranking and discovery system using PyTorch and LightFM, integrating continuous training pipelines and evaluation datasets via MLflow.

Zulip Open-Source GenAI Extensions

- **Scalable AI Systems:** Built scalable data pipelines and robust RAG systems using Node.js and Redis, heavily optimizing AI infrastructure to rapidly deliver low-latency NLP chat summaries.